

Discussion 1 Worksheet

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Problem 1: Determine which of the following are considered mathematical statements. If applicable, determine their truth values.

(i) $2 + 2 = 4$

(ii) $2 + 2 = 5$

(iii) $2 + 2$

(iv) $x^2 = 2$

(v) $\exists x \in \mathbb{Z} : x > 3$

(vi) $\forall x \in \mathbb{N} : x > -1$

(vii) 1,000,000 is a large number

Problem 2: Translate the following into plain English and determine their truth values.

(i) $\exists x \in \mathbb{Z} : x < 0$

(ii) $\exists x \in \mathbb{N} : x < 0$

(iii) $\forall x \in \mathbb{N}, \exists y \in \mathbb{R} : y > x$

Problem 3: Rewrite the following expressions using mathematical symbols.

(i) All natural numbers are greater than -1 .

(ii) π is not an integer.

(iii) A natural number multiplied by two is a natural number.

(iv) x is a rational number if it can be written as the quotient of two integers with a nonzero denominator.

(v) Between every two distinct real numbers, there is a rational number.

Problem 4: Determine the following truth values.

(i) $(2^3 = 9) \vee (3^2 = 9)$

(ii) $(2 + 2 = 5) \vee (2 * 3 = 5)$

(iii) $(\sqrt{4} \in \mathbb{N}) \wedge (\sqrt{2} \in \mathbb{Q})$

(iv) $(\sqrt{4} \in \mathbb{N}) \wedge (\sqrt{4} \in \mathbb{Z}) \wedge (\sqrt{4} \in \mathbb{Q}) \wedge (\sqrt{4} \in \mathbb{R})$

(v) $x < 2$

(vi) $(\{1, 2, 3, 4, 5, 6\} \subseteq \mathbb{N}) \wedge (\mathbb{N} \subseteq \mathbb{R})$